

ANIRBAN ROUTH

FULL-STACK SOFTWARE ENGINEER | MCA CANDIDATE | CLOUD & AI SPECIALIST

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TECHNICAL SUMMARY

Highly skilled Software Engineer and MCA student with extensive experience in **Java development, the MERN stack, and Cloud Architecture**. Expert in building and deploying scalable AI-driven applications using **Generative AI, Agentic AI, and RAG**. Proven track record in developing secure financial systems, NLP-based resume analysis platforms, and real-time collaborative marketplaces. Specialized in **AWS and Google Cloud Platform** deployment with 13+ professional certifications.

Languages: Java (JDK 11/17), Python, JavaScript (ES6+), SQL, HTML5/CSS3

AI/ML & NLP: RAG, KNN, TF-IDF, Hugging Face, Groq API, Agentic AI

Databases: MongoDB Atlas, PostgreSQL, MySQL, Redis (Caching)

Frameworks: React.js, Node.js, Express.js, Tailwind CSS, Mongoose, JWT

Cloud/DevOps: AWS (EC2, S3), GCP, Git, GitHub, Vercel, Cloud SQL

Security: 2FA, OAuth 2.0, Socket.io, Postman, Advanced Excel

PROFESSIONAL EXPERIENCE

Business Processing Associate — Tata Consultancy Services (TCS)

July 2023 – Sept 2025

- Engineered automated reporting systems for large-scale datasets using **Advanced Excel and VBA**, reducing manual processing time by 35% while maintaining a 99% accuracy rate for global clients.
- Collaborated with data teams to visualize complex business metrics, transforming raw data into high-impact executive dashboards that directly supported strategic investment and operational decisions.

IT Web Developer Intern — Royal Research

May 2023 – July 2023

- Spearheaded the debugging of 100+ critical **Java and Web** application issues; developed 50+ automated test cases using **Python and Java** to ensure robust performance across academic IT platforms.
- Maintained high standards for technical documentation, collaborating with senior developers to validate system requirements and architectural blueprints for complex software modules.

SELECTED TECHNICAL PROJECTS

MutualFund-Sahai-Hai — Final Year Project

MERN | AWS | 2FA | AI INTEGRATION

- Designed a secure, full-stack financial platform with **Two-Step Authentication (2FA)** and Google Backend integration. Deployed on **AWS EC2**, leveraging AI to provide personalized fund insights and risk analysis based on real-time market trends and historical performance metrics.

Resume IQ — AI-Based Analysis Platform (NEAT Minor Project)

PYTHON | KNN | TF-IDF | HUGGING FACE

- Built an NLP-powered engine using **TF-IDF vectorization** and **KNN algorithms** to classify resumes into 25+ domains with high precision. Integrated Hugging Face models for feature extraction to simulate real-world ATS scoring, helping candidates optimize profiles for specific Job Descriptions.

AI-Powered EdTech Ecosystem (NextStepper)

MERN | CLOUD SQL | GEN-AI | GCP

- Architecting a learning platform utilizing **Agentic AI** to automate career pathing. Implementing a hybrid cloud strategy with MongoDB for content and **GCP Cloud SQL** for transactional integrity.

Agentic AI Document Assistant & SkillBridge Marketplace

GROQ API | RAG | SOCKET.IO | RAZORPAY

- Built a **Context-Aware RAG Agent** capable of processing 200MB+ documents with 98% accuracy. Developed **SkillBridge**, a marketplace with real-time messaging via Socket.io and secure payment simulations.

EDUCATION & CERTIFICATIONS

Master of Computer Applications (MCA) — Netaji Subhash Engineering College

2024 – Present

Bachelor of Computer Applications (BCA) — NSEC (GPA: 8.7 / 10)

2020 – 2023

- Cybersecurity Workshop (ISOEH):** Specialized in Network Security, Ethical Hacking, and Pentesting (Feb 2026).
- Google Cloud:** 13 Professional Skill Badges including Data Analytics, GenAI, and Cloud Infrastructure.
- AI/ML Minor Project Cert:** Issued by N E A T for implementing high-accuracy NLP analysis systems (Dec 2025).

Languages: English (Fluent), Hindi, Bengali
Profiles: Leetcode (Java/Python Proficiency)

Academic: Class XII (65%) | Class X (77.4%)
Location: Kolkata, West Bengal